

IEEE 14-Bus GFL↔GFM Switching with Project AGSI++

Production all-KCL 160-s switching and SG-reference handback

4 August 2026

Abstract

One six-state EMF6 synchronous generator (SG) and four registered full-state dual-mode IBRs are simulated by the project-owned fixed-step, all-KCL hybrid engine for 160 s. The exact chronology is SG trip at 20 s, load increase at 50 s, bus-9 fault at 85–85.15 s, line 6–13 trip at 110 s, topology restoration and SG-close request at 145 s, and endpoint at 160 s. The run converges with maximum accepted-step residual 9.98×10^{-9} and at most one subdivision level. The synchronism guard accepts the SG at 147.175 s; all IBRs then return from GFM to GFL through the separately declared coordinated SG-reference handback. Final SG frequency is 60.0468 Hz and is still settling. No Bayesian-optimisation controller is used. A seeded band-limited display ripple is drawn beside, not instead of, each raw trace and affects figures only.

1 Case, decision index, and switching semantics

The system base is 100 MVA and 60 Hz. The SG is at bus 1; IBR1–IBR4 are at buses 2, 3, 6, and 8. The project AGSI++ index is

$$J_V = \frac{|V - V_{ref}|}{0.10}, \quad J_f = \frac{|f - 60|}{0.50}, \quad J_R = \frac{|df/dt|}{1.0}, \quad (1)$$

$$J_P = \frac{|P_{ref} - P|}{0.20}, \quad J_{SCR} = \max(0, 3/SCR - 1), \quad J_{lock} = \frac{|v_q|}{0.10}, \quad (2)$$

$$J_{GRA} = 1 - GRA, \quad AGSI_i^{++} = \min\left(1, \max\left(0, \frac{1}{7} \sum J_k\right)\right). \quad (3)$$

Here $GRA = 1$ when an online SG or committed GFM supplies a reference. The generic local state machine uses $\Gamma_{on} = 0.65$, $\Gamma_{off} = 0.35$, $T_{d,on} = 0.10$ s, and $T_{d,off} = 1.00$ s. In this declared chronology, however, the SG-trip transaction is a CASE_DEFINED all-GFM override, and the return at SG close is a coordinated handback. Neither transaction is presented as an index-threshold crossing.

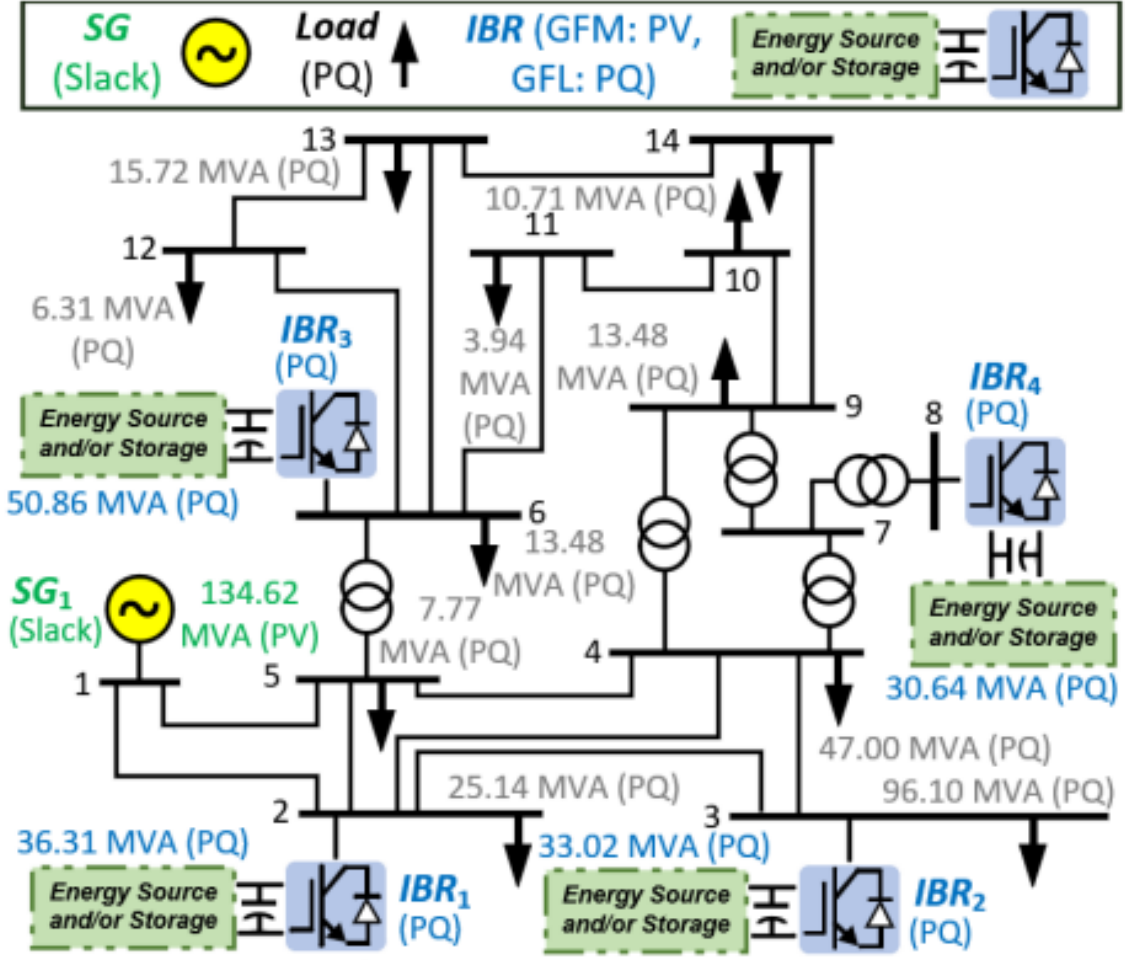


Figure 1: Network and resource locations used by the simulation.

2 Dynamic equations

The operational SG state is $x_{SG} = [\delta, \Delta\omega, E'_q, E'_d, E''_q, E''_d]^T$:

$$\dot{\delta} = \omega_0 \Delta\omega, \quad 2H\dot{\Delta\omega} = P_m - P_e - D\Delta\omega, \quad (4)$$

$$T'_{d0}\dot{E}'_q = E_{fd} + c_d E''_q - d_d E'_q, \quad T'_{q0}\dot{E}'_d = c_q E''_d - d_q E'_d, \quad (5)$$

$$T''_{d0}\dot{E}''_q = E'_q - E''_q - (X'_d - X''_d)I_d, \quad T''_{q0}\dot{E}''_d = E'_d - E''_d + (X'_q - X''_q)I_q. \quad (6)$$

The source singular limit $T'_{q0} = 0$ freezes $E'_d = 0$ exactly. Two prime-mover states remain outside the six-state electromagnetic order:

$$T_{SV}\dot{P}_{SV} = -P_{SV} + P_c - \Delta\omega/R, \quad T_{CH}\dot{P}_m = -P_m + P_{SV}.$$

They provide offline runback/phase preparation and post-close primary regulation. The numerical mapping is diagnostic and is not a machine-specific governor validation.

Each IBR retains the explicit 20-state switching superset

$$x_{GFM} = [\Delta\omega_m, \delta_{IT}, x_w, x_E, \delta_{PLL}, x_{PLL}, P_{inv,f}, I_{d,f}, Q_{inv,f}, V_{inv,f}, I_{q,f}, \delta_{IT,max}, \delta_{IT,min}]^T, \quad (7)$$

$$x_{GFL} = [V_{t,f}, P_f, I_{q,cmd,f}, P_{ord}, V_{LVPL,f}, I_{p,reg}, I_{q,reg}]^T. \quad (8)$$

The GFM branch contains the sourced VSM swing, PLL, voltage integrator, five measurement filters, and dynamic current-angle bounds; for example

$$2H\dot{\Delta\omega}_m = P_{ref,inv} - P_{inv,f} - (1/m_p + D_1)\Delta\omega_m - D_2(\Delta\omega_m - x_w), \quad (9)$$

$$T_{pf}\dot{P}_{inv,f} = \kappa P - P_{inv,f}, \quad T_{Vf}\dot{V}_{inv,f} = |V| - V_{inv,f}. \quad (10)$$

The GFL branch uses the registered WECC REGC_A and REEC_A models. Its states represent voltage and power filters, power order, LVPL, and regulated-current dynamics. Only the active branch enters the dynamic solve. Atomic state mapping is followed by a full-KCL right-limit solve; a failed solve rolls the transaction back.

3 Power-flow operating point

Table 1: Bus power-flow results (pu)

Bus	Type	$ V $	θ (deg)	P_g	Q_g	P_L	Q_L
1	REF	1.06000	0.0000	1.31895	-0.26951	0.00000	0.00000
2	PQ	1.05456	-2.9058	0.31859	0.17419	0.21700	0.12700
3	PQ	1.03301	-7.6873	0.28972	0.15841	0.94200	0.19000
4	PQ	1.04895	-5.9618	0.00000	0.00000	0.47800	-0.03900
5	PQ	1.05072	-4.9102	0.00000	0.00000	0.07600	0.01600
6	PQ	1.12705	-5.9341	0.44625	0.24399	0.11200	0.07500
7	PQ	1.09514	-6.1353	0.00000	0.00000	0.00000	0.00000
8	PQ	1.11749	-3.9176	0.26884	0.14699	0.00000	0.00000
9	PQ	1.09398	-7.6393	0.00000	0.00000	0.29500	0.16600
10	PQ	1.09237	-7.5903	0.00000	0.00000	0.09000	0.05800
11	PQ	1.10580	-6.8756	0.00000	0.00000	0.03500	0.01800
12	PQ	1.11174	-6.7892	0.00000	0.00000	0.06100	0.01600
13	PQ	1.10530	-6.9206	0.00000	0.00000	0.13500	0.05800
14	PQ	1.08161	-8.2396	0.00000	0.00000	0.14900	0.05000

Table 2: From-end line flows and losses (pu)

Line	From	To	P_{from}	Q_{from}	P_{loss}	Q_{loss}
1	1	2	0.900928	-0.203057	0.014518	-0.014696
2	1	5	0.418019	-0.066451	0.008475	-0.019813
3	2	3	0.464318	-0.000621	0.009133	-0.009246
4	2	4	0.314334	-0.079982	0.005358	-0.021354
5	2	5	0.209348	-0.060569	0.002332	-0.031219
6	3	4	-0.197092	-0.022968	0.002456	-0.007604
7	4	5	-0.447907	0.102332	0.002561	0.008079
8	4	7	0.017005	-0.115901	0.000000	0.002495
9	4	9	0.062330	-0.021423	0.000000	0.002062
10	5	6	0.090091	0.002266	0.000000	0.001610
11	6	11	0.130988	0.058687	0.001540	0.003226
12	6	12	0.085939	0.026725	0.000784	0.001631
13	6	13	0.207419	0.084233	0.002610	0.005140
14	7	8	-0.268840	-0.133746	0.000000	0.013242
15	7	9	0.285846	0.015351	0.000000	0.007516
16	9	10	-0.003741	0.022196	0.000013	0.000036
17	9	14	0.056917	0.023544	0.000403	0.000857
18	10	11	-0.093754	-0.035839	0.000693	0.001622
19	12	13	0.024155	0.009094	0.000119	0.000108
20	13	14	0.093845	0.030080	0.001359	0.002767

4 160-s event and mode evidence

Table 3: Accepted event transactions; AGSI++ is reported but overrides are explicit.

Time (s)	Event / index	Timer	Mode before → after	Reason
20.000	SG trip / 0.067	reference-loss guard	all GFL → all GFM	SG opened; atomic all-GFM reference-security commit
50.000	load step / 0.485	–	all GFM → all GFM	base constant-impedance load increased
85.000	fault on / 0.616	–	all GFM → all GFM	bus-9 shunt fault applied
85.150	fault clear / 1.000	–	all GFM → all GFM	fault admittance removed
110.000	line trip / 0.616	–	all GFM → all GFM	line 6–13 opened
145.000	topology restore / 0.609	–	all GFM → all GFM	base load and line restored
145.000	SG close request / 0.609	–	all GFM → all GFM	earliest close request; guard monitored
147.175	SG reclose / 0.485	0.5-s dwell	all GFM → all GFL	guard passed; SG reference hand-back

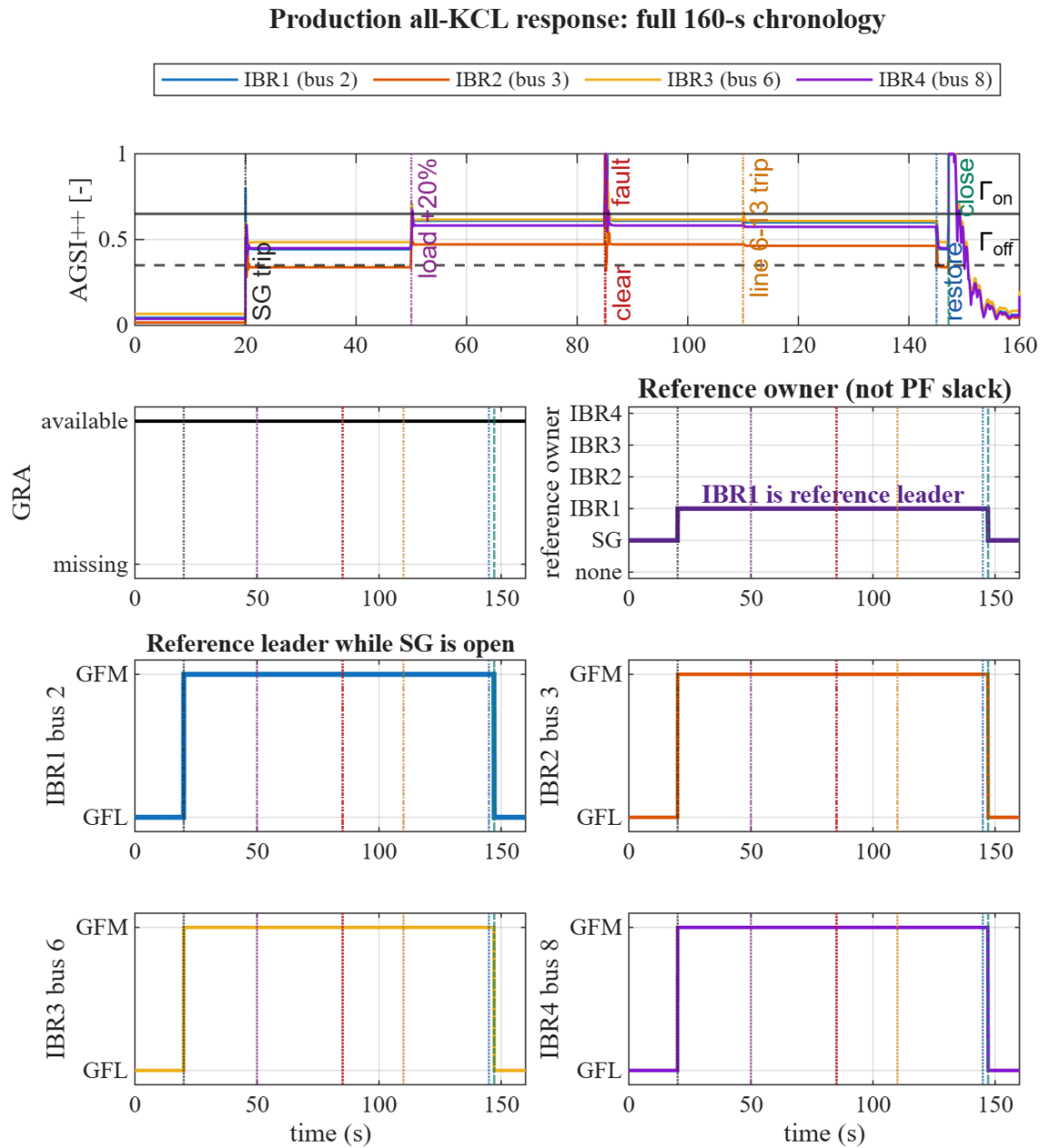


Figure 2: Bounded AGSI++, GRA, reference ownership, and separate IBR1–IBR4 mode timelines. The owner trace identifies IBR1 at bus 2 as the grid-forming angle/frequency leader while the SG is open. All IBRs are GFL before 20 s, GFM until the accepted handback, and GFL thereafter.

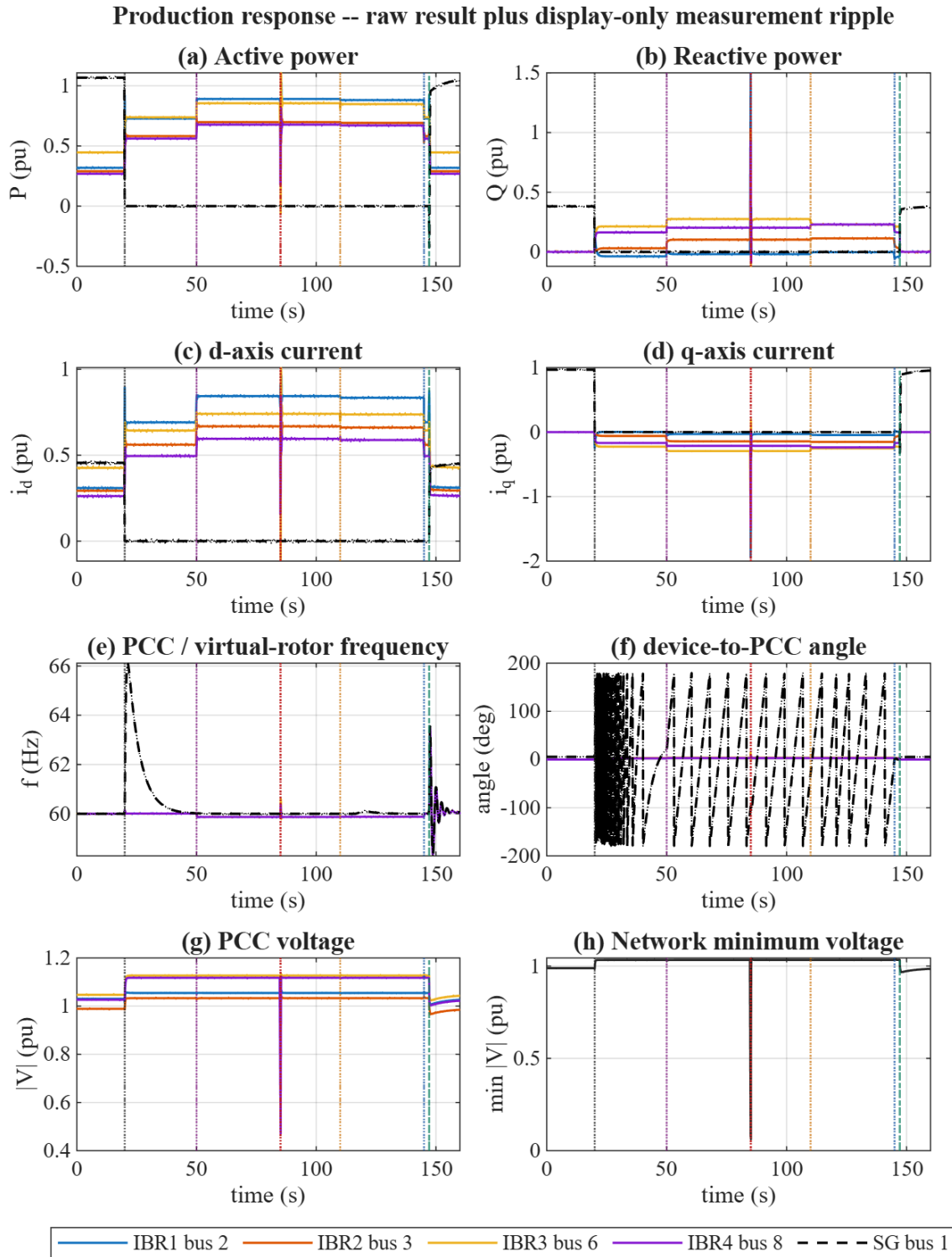


Figure 3: IBR and SG active/reactive power, frame currents, frequency, device-to-PCC angle, PCC voltage, and network minimum voltage. Solid traces are raw production results; dotted band-limited ripple is display-only and does not enter any equation or decision. Remaining vertical event excursions belong to the raw switching transient, not to the ripple.

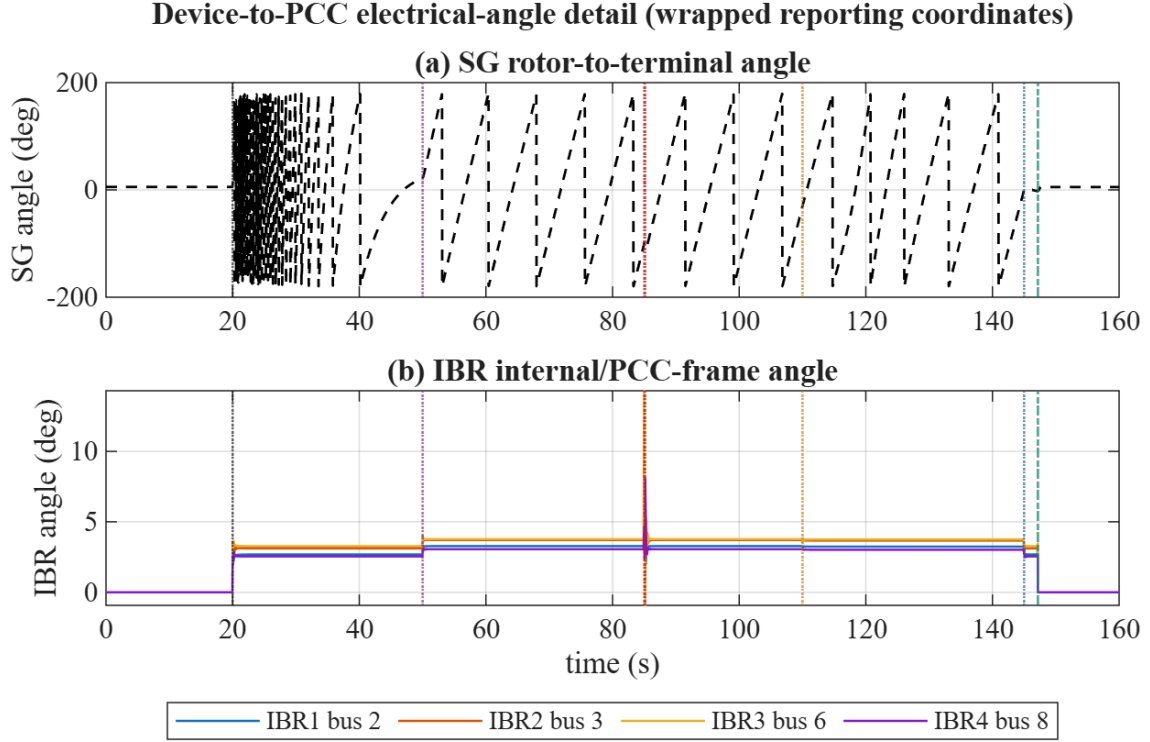


Figure 4: Angle detail. The offline SG rotor slips relative to the energized grid; the plotted coordinate is wrapped to $[-180, 180)$ degrees.

At the accepted close, the synchronism mismatch is $\Delta V = 0.01277$ pu, $\Delta f = 7.796 \times 10^{-4}$ pu, and $\Delta \theta = 3.402^\circ$. At 160 s the bus-voltage range is 0.9849–1.0556 pu, SG electrical power is 1.0482 pu, mechanical input is 1.0462 pu, and SG frequency is 60.0468 Hz with a nonzero settling slope. Thus the trajectory demonstrates a successful declared handback, but not exact steady-state restoration at the endpoint.

5 Evidence boundary and reproduction

The primary run uses $\Delta t = 0.0125$ s. A $\Delta t = 0.025$ s refinement comparison closes at 148.175 s rather than 147.175 s, so event-time independence is not claimed. The run proves numerical execution of the declared chronology, atomic mode transfer, full KCL, synchronism-gated close, and return to all GFL. It does not validate the diagnostic synchronizer, governor parameters, energy reserve, or a general protection design against field measurements. The dotted display ripple is never used as model evidence; the solid raw traces remain visible.

```
pf_init_paths;
addpath(fullfile('scripts', 'reporting'));
generate_ieee14_switch_report_figures(reuse_cache=true);
```

All production solvers are project-owned MATLAB. The targeted numerical, producer and consumer, hybrid event, fail-closed, and external-solver-prohibition suites pass 108/108 tests. The full repository regression was intentionally omitted under the repository risk policy because these targeted gates cover the changed producer, consumers, and failure paths.